Lecture 17 (10/23)

More examples of multiple-predictor regression

Midterm 2 material overview (Units 3 and 4)

Unit 3: estimation and single-predictor regression

- Estimators vs. parameters; Sample distribution.
- Bias, variance, MSE, consistency; standard errors and confidence intervals.
- Least-squares slope and intercept; Sample distribution of slope; R^2 and p-values (their relation to sample dist.)
- AR processes: stationarity, lagged regression, autocorrelation.

Unit 4: multiple-predictor regression

- Randomized trials vs. observational studies; confounding variables.
- Interpret coefficients in multiple predictor model
- Derive and apply omitted variable bias for one or many omitted predictors; Simpson's paradox.
- Categorical predictors and dummy variables.
- Effects on sample distribution of adding predictors.

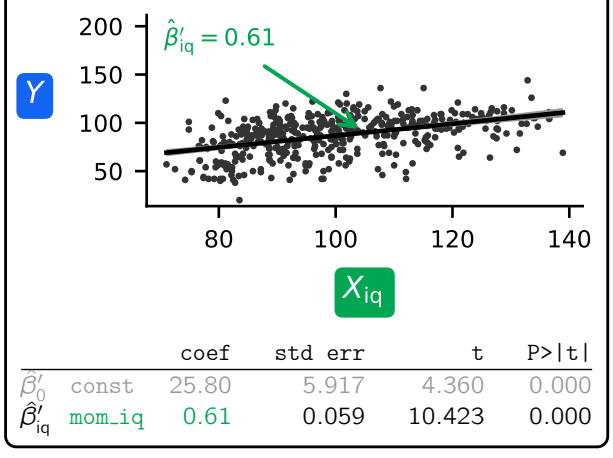
Today: Focus on applied examples of the two-predictor model.

Monday: Focus on derivations.

i	Y_i	$X_{\text{iq},i}$	$X_{\text{hs},i}$
	kid_score	mom_iq	mom_hs
1	65	121.1	1
2	98	89.4	1
⋮	⋮	⋮	⋮
6	98	107.9	0
⋮	⋮	⋮	⋮
N	70	91.3	1

i	Y_i	$X_{\text{iq},i}$	$X_{\text{hs},i}$
	kid_score	mom_iq	mom_hs
1	65	121.1	1
2	98	89.4	1
⋮	⋮	⋮	⋮
6	98	107.9	0
⋮	⋮	⋮	⋮
N	70	91.3	1

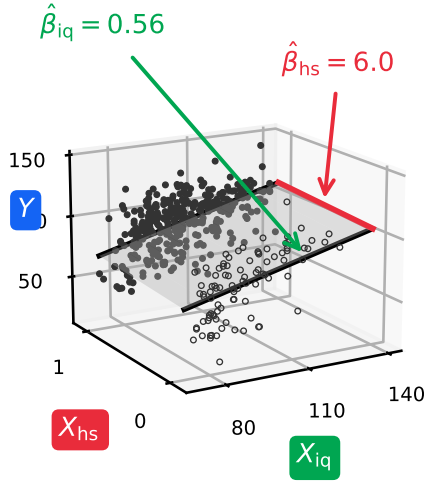
$$Y = \beta'_0 + \beta'_{\text{iq}} X_{\text{iq}} + \epsilon'$$

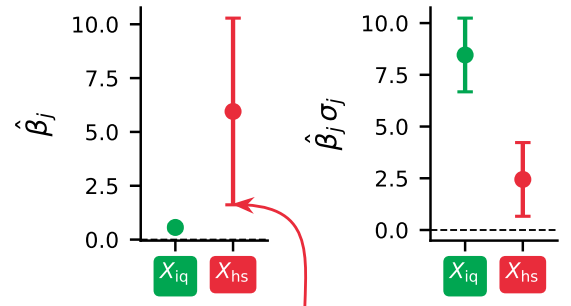


		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	25.80	5.917	4.360	0.000
$\hat{\beta}'_{\text{iq}}$	mom_iq	0.61	0.059	10.423	0.000

i	Y_i	$X_{\text{iq},i}$	$X_{\text{hs},i}$
	kid_score	mom_iq	mom_hs
1	65	121.1	1
2	98	89.4	1
⋮	⋮	⋮	⋮
6	98	107.9	0
⋮	⋮	⋮	⋮
N	70	91.3	1

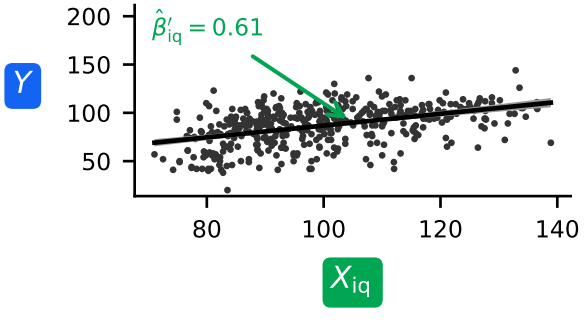
$$Y = \beta_0 + \beta_{\text{iq}} X_{\text{iq}} + \beta_{\text{hs}} X_{\text{hs}} + \epsilon$$





		coef $\hat{\theta}$	std err se($\hat{\theta}$)	t $\hat{T}$	P>|t| p_v
$\hat{\beta}_0$	const	25.73	5.875	4.380	0.000
$\hat{\beta}_{\text{iq}}$	mom_iq	0.56	0.061	9.309	0.000
$\hat{\beta}_{\text{hs}}$	mom_hs	5.95	2.212	2.690	0.007

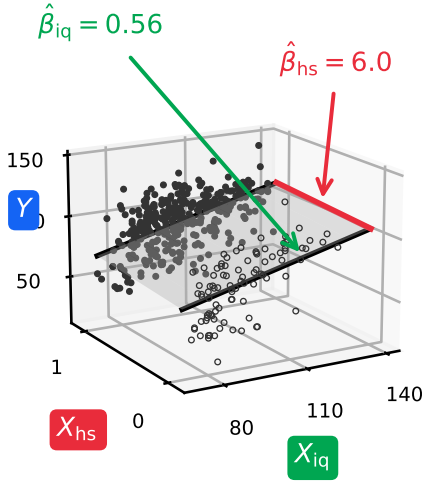
$$Y = \beta'_0 + \beta'_{\text{iq}} X_{\text{iq}} + \epsilon'$$

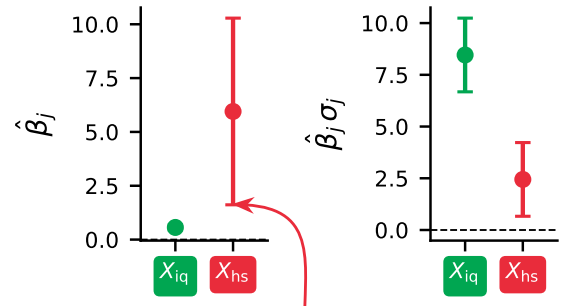


		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	25.80	5.917	4.360	0.000
$\hat{\beta}'_{\text{iq}}$	mom_iq	0.61	0.059	10.423	0.000

i	Y_i kid_score	$X_{\text{iq},i}$ mom_iq	$X_{\text{hs},i}$ mom_hs
1	65	121.1	1
2	98	89.4	1
⋮	⋮	⋮	⋮
6	98	107.9	0
⋮	⋮	⋮	⋮
N	70	91.3	1

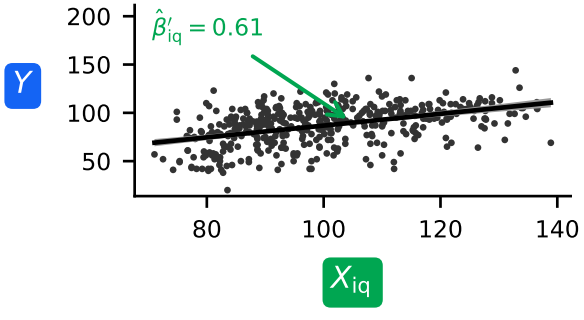
$$Y = \beta_0 + \beta_{\text{iq}} X_{\text{iq}} + \beta_{\text{hs}} X_{\text{hs}} + \epsilon$$





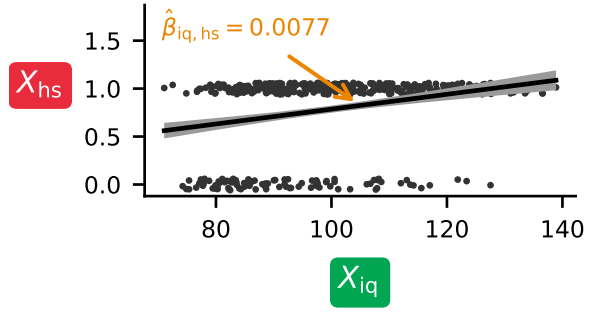
		coef $\hat{\theta}$	std err se($\hat{\theta}$)	t $\hat{T}$	P>|t| p_v
$\hat{\beta}_0$	const	25.73	5.875	4.380	0.000
$\hat{\beta}_{\text{iq}}$	mom_iq	0.56	0.061	9.309	0.000
$\hat{\beta}_{\text{hs}}$	mom_hs	5.95	2.212	2.690	0.007

$$Y = \beta'_0 + \beta'_{\text{iq}} X_{\text{iq}} + \epsilon'$$



		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	25.80	5.917	4.360	0.000
$\hat{\beta}'_{\text{iq}}$	mom_iq	0.61	0.059	10.423	0.000

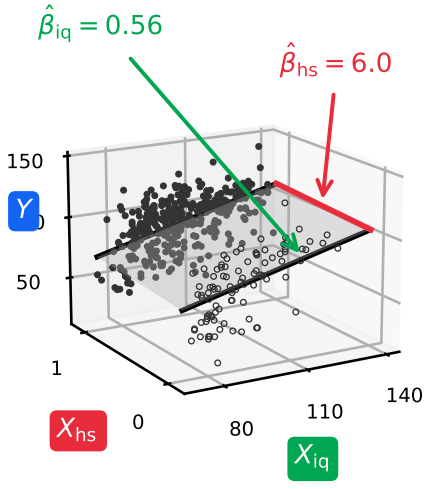
$$X_{\text{hs}} = \gamma_0 + \beta_{\text{iq,hs}} X_{\text{iq}} + \delta$$

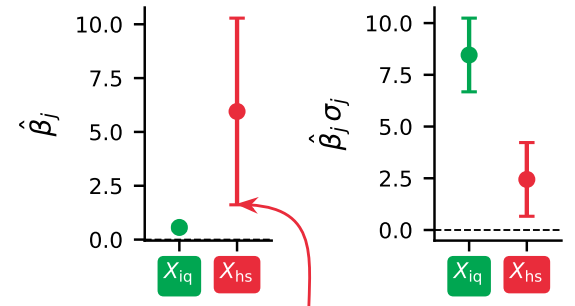


		coef	std err	t	P>|t|
$\hat{\gamma}_0$	const	0.0115	0.128	0.090	0.929
$\hat{\beta}_{\text{iq,hs}}$	mom_iq	0.0077	0.001	6.126	0.000

i	Y_i kid_score	$X_{\text{iq},i}$ mom_iq	$X_{\text{hs},i}$ mom_hs
1	65	121.1	1
2	98	89.4	1
⋮	⋮	⋮	⋮
6	98	107.9	0
⋮	⋮	⋮	⋮
N	70	91.3	1

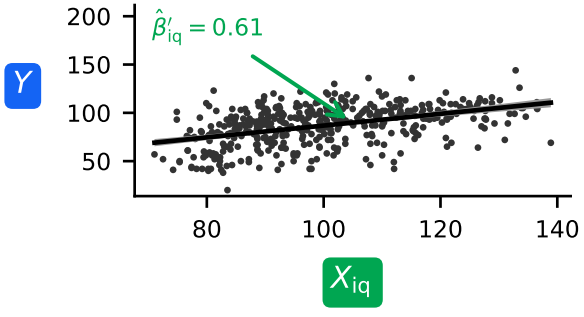
$$Y = \beta_0 + \beta_{\text{iq}} X_{\text{iq}} + \beta_{\text{hs}} X_{\text{hs}} + \epsilon$$





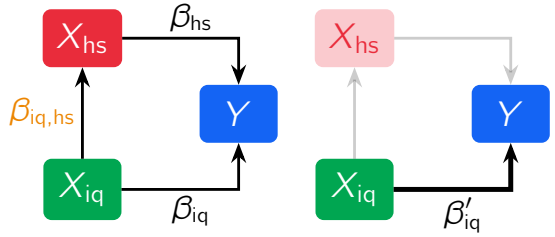
		coef $\hat{\theta}$	std err se($\hat{\theta}$)	t $\hat{T}$	P>|t| p_v
$\hat{\beta}_0$	const	25.73	5.875	4.380	0.000
$\hat{\beta}_{\text{iq}}$	mom_iq	0.56	0.061	9.309	0.000
$\hat{\beta}_{\text{hs}}$	mom_hs	5.95	2.212	2.690	0.007

$$Y = \beta'_0 + \beta'_{\text{iq}} X_{\text{iq}} + \epsilon'$$



		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	25.80	5.917	4.360	0.000
$\hat{\beta}'_{\text{iq}}$	mom_iq	0.61	0.059	10.423	0.000

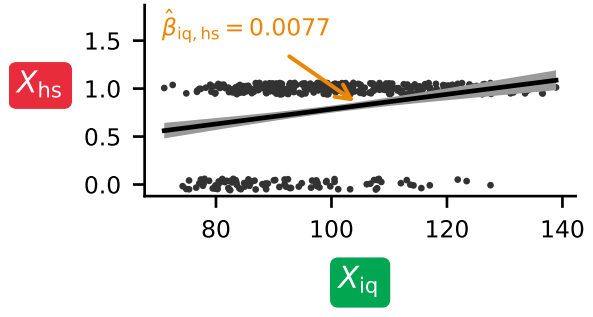
$$\beta'_{\text{iq}} = \beta_{\text{iq}} + \beta_{\text{hs}} \beta_{\text{iq,hs}}$$



both X_{hs} left out

omitted variable bias

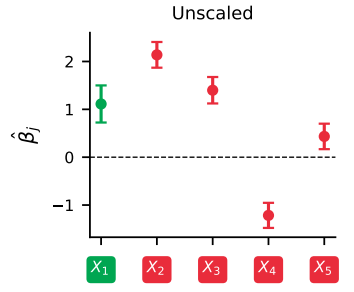
$$X_{\text{hs}} = \gamma_0 + \beta_{\text{iq,hs}} X_{\text{iq}} + \delta$$

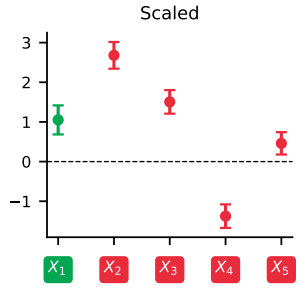


		coef	std err	t	P>|t|
$\hat{\gamma}_0$	const	0.0115	0.128	0.090	0.929
$\hat{\beta}_{\text{iq,hs}}$	mom_iq	0.0077	0.001	6.126	0.000

i	Y_i	$X_{1,i}$	$X_{2,i}$	$X_{3,i}$	$X_{4,i}$	$X_{5,i}$
1	6.7	1.7	3.0	0.3	1.1	0.4
2	6.0	−0.3	1.5	0.7	0.8	−0.9
3	−4.7	−0.5	−2.0	0.3	−1.1	−1.4
⋮	⋮	⋮	⋮	⋮	⋮	⋮
N	6.6	0.4	0.8	0.2	−1.9	1.0

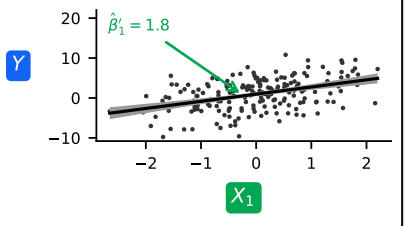
$$Y = \beta_0 + \beta_1 X_1 + \sum_{j=2}^{5} \beta_j X_j + \epsilon$$





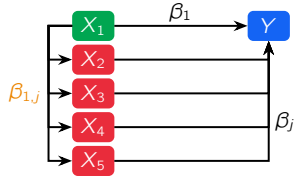
		coef $\hat{\theta}$	std err se($\hat{\theta}$)	t $\hat{T}$	P>|t| p_v
$\hat{\beta}_0$	const	1.01	0.137	7.370	0.000
$\hat{\beta}_1$	x1	1.11	0.198	5.622	0.000
$\hat{\beta}_2$	x2	2.14	0.137	15.604	0.000
$\hat{\beta}_3$	x3	1.40	0.141	9.925	0.000
$\hat{\beta}_4$	x4	−1.22	0.134	−9.090	0.000
$\hat{\beta}_5$	x5	0.43	0.136	3.195	0.002

$$Y = \beta'_0 + \beta'_1 X_1 + \epsilon'$$

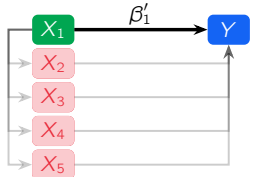


		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	0.96	0.251	3.807	0.000
$\hat{\beta}'_1$	x1	1.79	0.267	6.731	0.000

$$\beta'_1 = \beta_1 + \sum_{j=2}^{5} \beta_j \beta_{1,j}$$



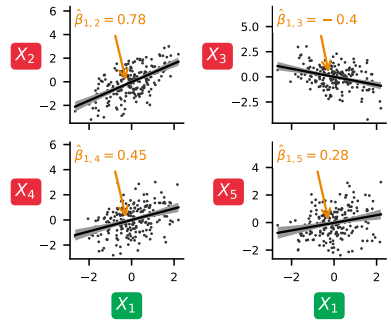
all predictors



$X_2, \ldots, X_5$ left out

omitted variable bias

$$X_j = \gamma_{0j} + \beta_{1,j} X_1 + \delta_j$$



		coef	std err	t	P>|t|
$\hat{\beta}_{1,2}$	x1	0.78	0.076	10.333	0.000
$\hat{\beta}_{1,3}$	x1	−0.40	0.076	−5.331	0.000
$\hat{\beta}_{1,4}$	x1	0.45	0.079	5.759	0.000
$\hat{\beta}_{1,5}$	x1	0.28	0.077	3.574	0.000

Example

Which hospital is better? A study follows 1000 patients, 500 at each hospital. Y is days until recovery. X_A is an indicator for hospital (1 if A, 0 if B). X_{sev} indicates severity (1 for a severe case, 0 otherwise).

Dep. Variable:		days	R-squared:		0.09
	coef	std err	t	P>|t|	[0.025	0.975]
const	7	0.21	32.97	0.00	6.58	7.42
hosp_A	3	0.30	9.99	0.00	2.41	3.59

Within either hospital, severe cases take 10 days longer to recover on average than mild cases. Severe cases make up 70% of hospital A's patients and 20% of hospital B's patients.

Question

Which hospital would you choose?

Solution: comparing hospitals

1) The unadjusted line is $E[Y \mid X_A] = 7 + \boxed{3}\, X_A$. The unadjusted difference is 3 days.

Solution: comparing hospitals

1) The unadjusted line is $E[Y \mid X_A] = 7 + \boxed{3}\,\boxed{X_A}$. The unadjusted difference is 3 days.
2) From the supplied information, the severity coefficient is $\boxed{\beta_{\text{sev}} = 10}$. The auxiliary slope is the difference in severe-case proportions:

$$\boxed{\beta_{A,\text{sev}}} = E[X_{\text{sev}} \mid X_A = 1] - E[X_{\text{sev}} \mid X_A = 0] = 0.70 - 0.20 = \boxed{0.50}.$$

Solution: comparing hospitals

1) The unadjusted line is $E[Y \mid X_A] = 7 + \boxed{3}\,\boxed{X_A}$. The unadjusted difference is 3 days.
2) From the supplied information, the severity coefficient is $\boxed{\beta_{\text{sev}} = 10}$. The auxiliary slope is the difference in severe-case proportions:

$$\boxed{\beta_{A,\text{sev}}} = E[X_{\text{sev}} \mid X_A = 1] - E[X_{\text{sev}} \mid X_A = 0] = 0.70 - 0.20 = \boxed{0.50}.$$

3) Apply OVB and solve for the adjusted hospital coefficient:

$$\boxed{\beta'_A} = \beta_A + \boxed{\beta_{\text{sev}}}\,\boxed{\beta_{A,\text{sev}}}, \qquad 3 = \beta_A + 10(0.5)$$

$$\boxed{\beta_A = 3 - 5 = -2}.$$

Solution: comparing hospitals

1) The unadjusted line is $E[Y \mid X_A] = 7 + \boxed{3}\,\boxed{X_A}$. The unadjusted difference is 3 days.
2) From the supplied information, the severity coefficient is $\boxed{\beta_{\text{sev}} = 10}$. The auxiliary slope is the difference in severe-case proportions:

$$\boxed{\beta_{A,\text{sev}}} = E[X_{\text{sev}} \mid X_A = 1] - E[X_{\text{sev}} \mid X_A = 0] = 0.70 - 0.20 = \boxed{0.50}.$$

3) Apply OVB and solve for the adjusted hospital coefficient:

$$\boxed{\beta'_A} = \beta_A + \boxed{\beta_{\text{sev}}}\,\boxed{\beta_{A,\text{sev}}}, \qquad 3 = \beta_A + 10(0.5)$$

$$\boxed{\beta_A = 3 - 5 = -2}.$$

4) Adjusted difference: -2 days. Choose A: predicted recovery is two days shorter at either severity.

Example

Is blood pressure related to age? Group A samples 400 people; Group B samples 1500 from the same population. Y is systolic blood pressure (mmHg), and X_{age} is age in decades. Only Group B also measures X_{bmi}, which is body mass index.

Group A

Dep. Variable:			bp	R-squared:		
	coef	std err	t	P>|t|	[0.025	0.975]
const	124	3.22	38.55	0.00	117.68	130.32
age	7	0.62	11.36	0.00	5.79	8.21

Group B

Dep. Variable:			bp	R-squared:		0.56
	coef	std err	t	P>|t|	[0.025	0.975]
const	90	1.62	55.42	0.00	86.81	93.19
age	3	0.27	11.12	0.00	2.47	3.53
bmi	2	0.06	33.16	0.00	1.88	2.12
Dep. Variable:			bmi	R-squared:		0.20
	coef	std err	t	P>|t|	[0.025	0.975]
const	17	0.54	31.51	0.00	15.94	18.06
age	2	0.10	19.35	0.00	1.80	2.20
Dep. Variable:			age	R-squared:		0.20
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.3	0.14	16.00	0.00	2.02	2.58
bmi	0.1	0.01	19.35	0.00	0.09	0.11

Question

Are the results of the two groups consistent?

Solution: blood pressure and omitted BMI

1) Group B estimates the adjusted relationship:

$$E[Y \mid X_{\text{age}}, X_{\text{bmi}}] = 90 + \boxed{3}\, X_{\text{age}} + \boxed{2}\, X_{\text{bmi}}.$$

Its auxiliary regression gives $E[X_{\text{bmi}} \mid X_{\text{age}}] = 17 + \boxed{2}\, X_{\text{age}}$.

Solution: blood pressure and omitted BMI

1) Group B estimates the adjusted relationship:

$$E[Y \mid X_{\text{age}}, X_{\text{bmi}}] = 90 + \boxed{3}\, X_{\text{age}} + \boxed{2}\, X_{\text{bmi}}.$$

   Its auxiliary regression gives $E[X_{\text{bmi}} \mid X_{\text{age}}] = 17 + \boxed{2}\, X_{\text{age}}$.

2) Omitting BMI changes the age slope by the product of two slopes:

$$\boxed{\hat{\beta}'_{\text{age}}} = \boxed{3} + \boxed{2} \times \boxed{2} = \boxed{7}.$$

   That agrees with Group A's fitted age slope.

Solution: blood pressure and omitted BMI

1) Group B estimates the adjusted relationship:

$$E[Y \mid X_{\text{age}}, X_{\text{bmi}}] = 90 + \boxed{3}\,\boxed{X_{\text{age}}} + \boxed{2}\,\boxed{X_{\text{bmi}}}.$$

Its auxiliary regression gives $E[X_{\text{bmi}} \mid X_{\text{age}}] = 17 + \boxed{2}\,\boxed{X_{\text{age}}}$.

2) Omitting BMI changes the age slope by the product of two slopes:

$$\boxed{\hat{\beta}'_{\text{age}}} = \boxed{3} + \boxed{2} \times \boxed{2} = \boxed{7}.$$

That agrees with Group A's fitted age slope.

3) The intercept also changes:

$$\hat{\beta}'_0 = 90 + 2 \times 17 = 124.$$

Thus Group B implies the reduced line $E[Y \mid X_{\text{age}}] = 124 + 7X_{\text{age}}$, exactly the reported Group A line (at the displayed precision).

Solution: blood pressure and omitted BMI

1) Group B estimates the adjusted relationship:

$$E[Y \mid X_{\text{age}}, X_{\text{bmi}}] = 90 + \boxed{3}\, X_{\text{age}} + \boxed{2}\, X_{\text{bmi}}.$$

   Its auxiliary regression gives $E[X_{\text{bmi}} \mid X_{\text{age}}] = 17 + \boxed{2}\, X_{\text{age}}$.

2) Omitting BMI changes the age slope by the product of two slopes:

$$\boxed{\hat{\beta}'_{\text{age}}} = \boxed{3} + \boxed{2} \times \boxed{2} = \boxed{7}.$$

   That agrees with Group A's fitted age slope.

3) The intercept also changes:

$$\hat{\beta}'_0 = 90 + 2 \times 17 = 124.$$

   Thus Group B implies the reduced line $E[Y \mid X_{\text{age}}] = 124 + 7X_{\text{age}}$, exactly the reported Group A line (at the displayed precision).

4) Compare with Group A: slope $7 = 7$ and intercept $124 = 124$. The fitted coefficients agree.

Example

Do books at home raise children's reading scores? A survey includes 600 children. Y is a reading test score, and X_{books} is the number of books at home, measured in tens.

Dep. Variable:		score	R-squared:		0.51
	coef	std err	t	P>|t|	[0.025	0.975]
const	42	2.04	20.57	0.00	37.99	46.01
books	8	0.32	24.78	0.00	7.37	8.63

Two possible confounders are X_{educ}, parent's years of education, and X_{inc}, household income in \$10k. Assume the following linear regression relationships hold in this population:

- among children with the same number of books and the same income, each extra year of parent's education goes with 2 more points on average;
- among children with the same number of books and the same parent's education, each extra \$10k of income goes with 1 more point on average;
- each extra 10 books goes with 1 more year of parent's education and \$20k more income on average.

Question

What is the coefficient on `books` in a model that also includes X_{educ} and X_{inc}?

Solution: books, education, and income

1) Translate the given effects into full-model coefficients:

$$E[Y \mid X_{\text{books}}, X_{\text{educ}}, X_{\text{inc}}] = \hat{\beta}_0 + \hat{\beta}_{\text{books}} X_{\text{books}} + \boxed{2} X_{\text{educ}} + \boxed{1} X_{\text{inc}}.$$

Income is measured in units of \$10,000.

Solution: books, education, and income

1) Translate the given effects into full-model coefficients:

$$E[Y \mid X_{\text{books}}, X_{\text{educ}}, X_{\text{inc}}] = \hat{\beta}_0 + \hat{\beta}_{\text{books}} X_{\text{books}} + \boxed{2} X_{\text{educ}} + \boxed{1} X_{\text{inc}}.$$

Income is measured in units of \$10,000.

2) Each additional ten books corresponds to one year of education and two income units, so the auxiliary slopes are

$$\boxed{\beta_{\text{books,educ}} = 1}, \qquad \boxed{\beta_{\text{books,inc}} = 2}.$$

Solution: books, education, and income

1) Translate the given effects into full-model coefficients:

$$E[Y \mid X_{\text{books}}, X_{\text{educ}}, X_{\text{inc}}] = \hat{\beta}_0 + \hat{\beta}_{\text{books}} X_{\text{books}} + \boxed{2} X_{\text{educ}} + \boxed{1} X_{\text{inc}}.$$

Income is measured in units of \$10,000.

2) Each additional ten books corresponds to one year of education and two income units, so the auxiliary slopes are

$$\boxed{\beta_{\text{books,educ}} = 1}, \qquad \boxed{\beta_{\text{books,inc}} = 2}.$$

3) Apply OVB with both omitted predictors:

$$\boxed{8} = \hat{\beta}_{\text{books}} + \boxed{2} \times \boxed{1} + \boxed{1} \times \boxed{2}.$$

Solution: books, education, and income

1) Translate the given effects into full-model coefficients:

$$E[Y \mid X_{\text{books}}, X_{\text{educ}}, X_{\text{inc}}] = \hat{\beta}_0 + \hat{\beta}_{\text{books}} X_{\text{books}} + \boxed{2} X_{\text{educ}} + \boxed{1} X_{\text{inc}}.$$

   Income is measured in units of \$10,000.

2) Each additional ten books corresponds to one year of education and two income units, so the auxiliary slopes are

$$\boxed{\beta_{\text{books,educ}} = 1}, \qquad \boxed{\beta_{\text{books,inc}} = 2}.$$

3) Apply OVB with both omitted predictors:

$$\boxed{8} = \hat{\beta}_{\text{books}} + \boxed{2} \times \boxed{1} + \boxed{1} \times \boxed{2}.$$

4) Solve: $\boxed{\hat{\beta}_{\text{books}} = 4}$.

Example

Does a new training program improve athletic performance? In an observational study, athletes who joined the program scored 6 points higher on average on a fitness test than athletes who did not join.

Y is the fitness score. X_{program} is an indicator for participation (1 for participants, 0 for nonparticipants). The groups also differed in their habits before the program began:

Average before the program	Participants	Nonparticipants
Sleep (hours per night)	8	7
Exercise (hours per week)	5	3

In a linear model including program participation, prior sleep, and prior exercise, the coefficients on sleep and exercise are 2 points per hour and 1 point per hour, respectively.

Question

What is the coefficient on program participation after controlling for prior sleep and exercise?

Solution: the training-program comparison

1) The raw difference is the single-predictor coefficient:

$$\boxed{\beta'_{\text{program}} = 6}.$$

Solution: the training-program comparison

1) The raw difference is the single-predictor coefficient:

$$\boxed{\beta'_{\text{program}} = 6}.$$

2) A regression on a binary indicator has slope equal to the difference in group averages. From the table,

$$\boxed{\beta_{\text{program,sleep}} = 8 - 7 = 1}, \qquad \boxed{\beta_{\text{program,exercise}} = 5 - 3 = 2}.$$

Solution: the training-program comparison

1) The raw difference is the single-predictor coefficient:

$$\boxed{\beta'_{\text{program}} = 6}.$$

2) A regression on a binary indicator has slope equal to the difference in group averages. From the table,

$$\boxed{\beta_{\text{program,sleep}} = 8 - 7 = 1}, \qquad \boxed{\beta_{\text{program,exercise}} = 5 - 3 = 2}.$$

3) The supplied full-model coefficients are 2 for sleep and 1 for exercise. Apply OVB:

$$\boxed{6} = \beta_{\text{program}} + \boxed{2} \times \boxed{1} + \boxed{1} \times \boxed{2}.$$

Solution: the training-program comparison

1) The raw difference is the single-predictor coefficient:

$$\boxed{\beta'_{\text{program}} = 6}.$$

2) A regression on a binary indicator has slope equal to the difference in group averages. From the table,

$$\boxed{\beta_{\text{program,sleep}} = 8 - 7 = 1}, \qquad \boxed{\beta_{\text{program,exercise}} = 5 - 3 = 2}.$$

3) The supplied full-model coefficients are 2 for sleep and 1 for exercise. Apply OVB:

$$\boxed{6} = \beta_{\text{program}} + \boxed{2} \times \boxed{1} + \boxed{1} \times \boxed{2}.$$

4) Solve: $\boxed{\beta_{\text{program}} = 6 - 2 - 2 = 2}$ points.

Example

What makes a wildfire spread? A fictional study includes 25 wildfires. Y is area burned in acres. X_{drought} is a drought index from 0 to 100, and X_{wind} is wind speed in mph.

Dep. Variable:		acres	R-squared:			0.80
	coef	std err	t	P>|t|	[0.025	0.975]
const	-220	163.46	-1.35	0.19	-558.99	118.99
drought	5	3.73	1.34	0.19	-2.74	12.74
wind	5	9.14	0.55	0.59	-13.95	23.95
Dep. Variable:		acres	R-squared:			0.80
	coef	std err	t	P>|t|	[0.025	0.975]
const	-305	50.01	-6.10	0.00	-408.46	-201.54
drought	7	0.73	9.53	0.00	5.48	8.52
Dep. Variable:		acres	R-squared:			0.78
	coef	std err	t	P>|t|	[0.025	0.975]
const	-2.4	19.10	-0.13	0.90	-41.91	37.11
wind	17	1.86	9.14	0.00	13.15	20.85

Question

Is area burned related to drought? To wind? Do the large p-values in the first model mean that neither matters?

Solution: wildfire predictors and uncertainty

1) Individually, drought and wind predict area burned: the simple-model slopes are $\boxed{7}$ and $\boxed{17}$, with small p-values.

Solution: wildfire predictors and uncertainty

1) Individually, drought and wind predict area burned: the simple-model slopes are $\boxed{7}$ and $\boxed{17}$, with small p-values.
2) In the joint model, the slopes are $\boxed{5}$ and $\boxed{5}$, but their CIs include zero. Neither adjusted coefficient is significant at the 5% level.

Solution: wildfire predictors and uncertainty

1) Individually, drought and wind predict area burned: the simple-model slopes are $\boxed{7}$ and $\boxed{17}$, with small p-values.
2) In the joint model, the slopes are $\boxed{5}$ and $\boxed{5}$, but their CIs include zero. Neither adjusted coefficient is significant at the 5% level.
3) Use the changes in slope to infer the auxiliary slopes:

$$7 = 5 + 5\beta_{\text{drought,wind}} \Rightarrow \boxed{\beta_{\text{drought,wind}} = 0.4},$$

$$17 = 5 + 5\beta_{\text{wind,drought}} \Rightarrow \boxed{\beta_{\text{wind,drought}} = 2.4}.$$

Solution: wildfire predictors and uncertainty

1) Individually, drought and wind predict area burned: the simple-model slopes are $\boxed{7}$ and $\boxed{17}$, with small p-values.
2) In the joint model, the slopes are $\boxed{5}$ and $\boxed{5}$, but their CIs include zero. Neither adjusted coefficient is significant at the 5% level.
3) Use the changes in slope to infer the auxiliary slopes:

$$7 = 5 + 5\beta_{\text{drought,wind}} \Rightarrow \boxed{\beta_{\text{drought,wind}} = 0.4},$$

$$17 = 5 + 5\beta_{\text{wind,drought}} \Rightarrow \boxed{\beta_{\text{wind,drought}} = 2.4}.$$

4) Their product is $\rho^2 = 0.4(2.4) = 0.96$, so $\rho \approx 0.98$. High collinearity explains the large joint-model SEs; the large p-values do not imply zero coefficients.

Regression in both directions

Suppose

$$Y = 2 + 3X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2)$$
$$X = 6 + 1/6\,Y + \eta, \quad \eta \sim \text{Normal}(0, \sigma_\eta^2)$$

where ϵ is independent of X and η is independent of Y.

Question

- What is the correlation coefficient between X and Y?
- What is the relationship between σ_ϵ and σ_η?

Solution: regression in both directions

1) The two single-predictor slopes satisfy

$$\boxed{3} = \frac{\text{cov}(X, Y)}{\text{var}(X)}, \qquad \boxed{1/6} = \frac{\text{cov}(X, Y)}{\text{var}(Y)}.$$

Solution: regression in both directions

1) The two single-predictor slopes satisfy

$$\boxed{3} = \frac{\mathsf{cov}(X, Y)}{\mathsf{var}(X)}, \qquad \boxed{1/6} = \frac{\mathsf{cov}(X, Y)}{\mathsf{var}(Y)}.$$

2) Multiply to eliminate the marginal variances:

$$\rho^2 = 3(1/6) = 1/2.$$

Both slopes are positive, so $\boxed{\rho = 1/\sqrt{2}}$.

Solution: regression in both directions

1) The two single-predictor slopes satisfy

$$\boxed{3} = \frac{\mathsf{cov}(X, Y)}{\mathsf{var}(X)}, \qquad \boxed{1/6} = \frac{\mathsf{cov}(X, Y)}{\mathsf{var}(Y)}.$$

2) Multiply to eliminate the marginal variances:

$$\rho^2 = 3(1/6) = 1/2.$$

Both slopes are positive, so $\boxed{\rho = 1/\sqrt{2}}$.

3) Divide the slope formulas to find $\mathsf{var}(Y)/\mathsf{var}(X) = 3/(1/6) = 18$.

Solution: regression in both directions

1) The two single-predictor slopes satisfy

$$\boxed{3} = \frac{\mathrm{cov}(X, Y)}{\mathrm{var}(X)}, \qquad \boxed{1/6} = \frac{\mathrm{cov}(X, Y)}{\mathrm{var}(Y)}.$$

2) Multiply to eliminate the marginal variances:

$$\rho^2 = 3(1/6) = 1/2.$$

Both slopes are positive, so $\boxed{\rho = 1/\sqrt{2}}$.

3) Divide the slope formulas to find $\mathrm{var}(Y)/\mathrm{var}(X) = 3/(1/6) = 18$.

4) In either direction, residual variance is $(1 - \rho^2)$ times response variance:

$$\sigma_\epsilon^2 = \tfrac{1}{2}\,\mathrm{var}(Y), \qquad \sigma_\eta^2 = \tfrac{1}{2}\,\mathrm{var}(X) \quad \Rightarrow \quad \boxed{\sigma_\epsilon = \sqrt{18}\,\sigma_\eta}.$$

Predicting a future price

Example

A commodity's daily price Y_t (dollars) follows the model

$$Y_{t+1} = 50 + 0.8(Y_t - 50) + \epsilon_{t+1}, \quad \epsilon_t \sim \text{Normal}(0, 1)$$

The noise standard deviation is $\sigma_\epsilon = 1$. The noise terms ϵ_t are iid and independent of past prices. Today's price is $Y_t = 60$.

Question

1) What are the average predicted prices tomorrow and two days from today?
2) What is the approximate conditional variance of the price 10 years from now, given today's price?

Solution: forecasting the price

1) Center the process at its long-run mean, $Z_t = Y_t - 50$:

$$Z_{t+1} = \boxed{0.8}\, Z_t + \epsilon_{t+1}, \qquad Z_t = 60 - 50 = 10.$$

Solution: forecasting the price

1) Center the process at its long-run mean, $Z_t = Y_t - 50$:

$$Z_{t+1} = \boxed{0.8}\, Z_t + \epsilon_{t+1}, \qquad Z_t = 60 - 50 = 10.$$

2) The future noise terms are independent and have mean zero, so

$$E[Y_{t+1} \mid Y_t = 60] = 50 + 0.8(10) = 58,$$

$$E[Y_{t+2} \mid Y_t = 60] = 50 + 0.8^2(10) = 56.4.$$

Solution: forecasting the price

1) Center the process at its long-run mean, $Z_t = Y_t - 50$:

$$Z_{t+1} = \boxed{0.8}\, Z_t + \epsilon_{t+1}, \qquad Z_t = 60 - 50 = 10.$$

2) The future noise terms are independent and have mean zero, so

$$E[Y_{t+1} \mid Y_t = 60] = 50 + 0.8(10) = 58,$$

$$E[Y_{t+2} \mid Y_t = 60] = 50 + 0.8^2(10) = 56.4.$$

3) Since $|\rho| = 0.8 < 1$, use the stationary variance formula:

$$\text{var}(Y) = \frac{\sigma_\epsilon^2}{1 - \rho^2} = \frac{1}{1 - 0.8^2}.$$

Solution: forecasting the price

1) Center the process at its long-run mean, $Z_t = Y_t - 50$:

$$Z_{t+1} = \boxed{0.8}\, Z_t + \epsilon_{t+1}, \qquad Z_t = 60 - 50 = 10.$$

2) The future noise terms are independent and have mean zero, so

$$E[Y_{t+1} \mid Y_t = 60] = 50 + 0.8(10) = 58,$$

$$E[Y_{t+2} \mid Y_t = 60] = 50 + 0.8^2(10) = 56.4.$$

3) Since $|\rho| = 0.8 < 1$, use the stationary variance formula:

$$\text{var}(Y) = \frac{\sigma_\epsilon^2}{1 - \rho^2} = \frac{1}{1 - 0.8^2}.$$

4) After 10 years, the variance is approximately the stationary variance:

$$\boxed{\text{var}(Y_{t+h} \mid Y_t) \approx \frac{1}{0.36} \approx 2.78 \text{ dollars}^2}.$$

Two groups, different choices of X

Example

Two research groups measure a sensor's output Y at different values of X. Each takes 100 measurements. The same model $Y = \beta_0 + \beta_1 X + \epsilon$ holds for both groups, with independent Normal errors of the same variance.

```
Group A: N = 100    R-squared = 0.140    Residual std dev = 5.00
                 coef    std err          t      P>|t|
const           10.00       0.50      20.00      0.000
X                2.00       0.50       4.00      0.000

Group B: N = 100    R-squared = 0.395    Residual std dev = 5.00
                 coef    std err          t      P>|t|
const           10.00       0.50      20.00      0.000
X                2.00       0.25       8.00      0.000
```

Question

Which group sampled a wider range of X? By what factor do their standard deviations of X differ?

Solution: the spread of the sampled predictor

1) Both groups have $N = 100$ and residual SD $s = 5$. Their slope SEs are $\boxed{0.50}$ (A) and $\boxed{0.25}$ (B).

Solution: the spread of the sampled predictor

1) Both groups have $N = 100$ and residual SD $s = 5$. Their slope SEs are $\boxed{0.50}$ (A) and $\boxed{0.25}$ (B).
2) For regression with an intercept, conditional on the sampled x_i,

$$\text{se}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad S_{xx} = \sum_i (x_i - \bar{x})^2.$$

Solution: the spread of the sampled predictor

1) Both groups have $N = 100$ and residual SD $s = 5$. Their slope SEs are $\boxed{0.50}$ (A) and $\boxed{0.25}$ (B).
2) For regression with an intercept, conditional on the sampled x_i,

$$\text{se}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad S_{xx} = \sum_i (x_i - \bar{x})^2.$$

3) Invert the formula using each group's output:

$$S_{xx,A} = (5/0.50)^2 = 100, \qquad S_{xx,B} = (5/0.25)^2 = 400.$$

Solution: the spread of the sampled predictor

1) Both groups have $N = 100$ and residual SD $s = 5$. Their slope SEs are $\boxed{0.50}$ (A) and $\boxed{0.25}$ (B).
2) For regression with an intercept, conditional on the sampled x_i,

$$\text{se}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad S_{xx} = \sum_i (x_i - \bar{x})^2.$$

3) Invert the formula using each group's output:

$$S_{xx,A} = (5/0.50)^2 = 100, \qquad S_{xx,B} = (5/0.25)^2 = 400.$$

4) Since sample SD is $s_X = \sqrt{S_{xx}/(N-1)}$ and N is equal,

$$\boxed{\frac{s_{X,B}}{s_{X,A}} = \sqrt{400/100} = 2}.$$

Group B's sampled X has twice the standard deviation.

Example

Suppose that the two-predictor model

$$Y = 2 + 3X_1 + 4X_2 + \epsilon, \qquad \epsilon \sim \text{Normal}(0, 4)$$

is accurate, with ϵ independent of the predictors. Both predictors are binary, with $P(X_1 = 1) = 1/2$ and

	$P(X_2 = 0 \mid X_1)$	$P(X_2 = 1 \mid X_1)$
$X_1 = 0$	0.75	0.25
$X_1 = 1$	0.25	0.75

A researcher observes only X_1 and Y and fits

$$Y = \beta_0' + \beta_1' X_1 + \epsilon'.$$

Question

- What is β_1'?
- Does the single-predictor model satisfy the linear regression assumptions?

Solution: omitting the binary predictor

1) Read the table: $E[X_2 \mid X_1 = 0] = 0.25$ and $E[X_2 \mid X_1 = 1] = 0.75$. Thus

$$E[\,X_2 \mid X_1\,] = 0.25 + \boxed{0.5}\, X_1.$$

Solution: omitting the binary predictor

1) Read the table: $E[X_2 \mid X_1 = 0] = 0.25$ and $E[X_2 \mid X_1 = 1] = 0.75$. Thus

$$E[\,X_2 \mid X_1\,] = 0.25 + \boxed{0.5}\, X_1.$$

2) Substitute into the conditional mean of Y:

$$E[Y \mid X_1] = 2 + 3X_1 + 4(0.25 + 0.5X_1) = 3 + 5X_1.$$

So $\boxed{\beta_0' = 3}$, $\boxed{\beta_1' = 5}$, and $\epsilon' = 4(X_2 - 0.25 - 0.5X_1) + \epsilon$.

Solution: omitting the binary predictor

1) Read the table: $E[X_2 \mid X_1 = 0] = 0.25$ and $E[X_2 \mid X_1 = 1] = 0.75$. Thus

$$E[\,X_2 \mid X_1\,] = 0.25 + \boxed{0.5}\, X_1.$$

2) Substitute into the conditional mean of Y:

$$E[Y \mid X_1] = 2 + 3X_1 + 4(0.25 + 0.5X_1) = 3 + 5X_1.$$

So $\boxed{\beta_0' = 3}$, $\boxed{\beta_1' = 5}$, and $\epsilon' = 4(X_2 - 0.25 - 0.5X_1) + \epsilon$.

3) $E[\epsilon' \mid X_1] = 0$ and $\mathbf{var}(\epsilon' \mid X_1) = 16(0.25)(0.75) + 4 = 7$ in both groups. The conditional mean is linear and conditional variance is constant.

Solution: omitting the binary predictor

1) Read the table: $E[X_2 \mid X_1 = 0] = 0.25$ and $E[X_2 \mid X_1 = 1] = 0.75$. Thus

$$E[\,X_2 \mid X_1\,] = 0.25 + \boxed{0.5}\, X_1.$$

2) Substitute into the conditional mean of Y:

$$E[Y \mid X_1] = 2 + 3X_1 + 4(0.25 + 0.5X_1) = 3 + 5X_1.$$

So $\boxed{\beta_0' = 3}$, $\boxed{\beta_1' = 5}$, and $\epsilon' = 4(X_2 - 0.25 - 0.5X_1) + \epsilon$.

3) $E[\epsilon' \mid X_1] = 0$ and $\mathbf{var}(\epsilon' \mid X_1) = 16(0.25)(0.75) + 4 = 7$ in both groups. The conditional mean is linear and conditional variance is constant.

4) The reduced-model noise ϵ' is not exogenous: it is not independent of X_1.

Dummy variables: delivery methods

Example

A company uses bicycles, vans, or drones for deliveries. Y is delivery time in minutes. X_{van} is 1 for a van delivery and 0 otherwise. X_{drone} is 1 for a drone delivery and 0 otherwise. Each delivery uses exactly one method.
The fitted model is

$$E[Y \mid X_{\text{van}}, X_{\text{drone}}] = 18 + 6X_{\text{van}} - 8X_{\text{drone}}.$$

Question

Interpret the three regression coefficients.

Solution: dummy variables

1) Bicycle is the baseline: $X_{\text{van}} = 0$ and $X_{\text{drone}} = 0$.

$$E[Y \mid X_{\text{van}} = 0, X_{\text{drone}} = 0] = \boxed{18} \text{ minutes.}$$

Solution: dummy variables

1) Bicycle is the baseline: $X_{\text{van}} = 0$ and $X_{\text{drone}} = 0$.

$$E[Y \mid X_{\text{van}} = 0, X_{\text{drone}} = 0] = \boxed{18} \text{ minutes.}$$

2) The van coefficient is $\boxed{6}$: van deliveries average 6 minutes longer than bicycle deliveries.

$$E[Y \mid X_{\text{van}} = 1, X_{\text{drone}} = 0] = 18 + 6(1) - 8(0) = 24 \text{ minutes.}$$

Solution: dummy variables

1) Bicycle is the baseline: $X_{\text{van}} = 0$ and $X_{\text{drone}} = 0$.

$$E[Y \mid X_{\text{van}} = 0, X_{\text{drone}} = 0] = \boxed{18} \text{ minutes.}$$

2) The van coefficient is $\boxed{6}$: van deliveries average 6 minutes longer than bicycle deliveries.

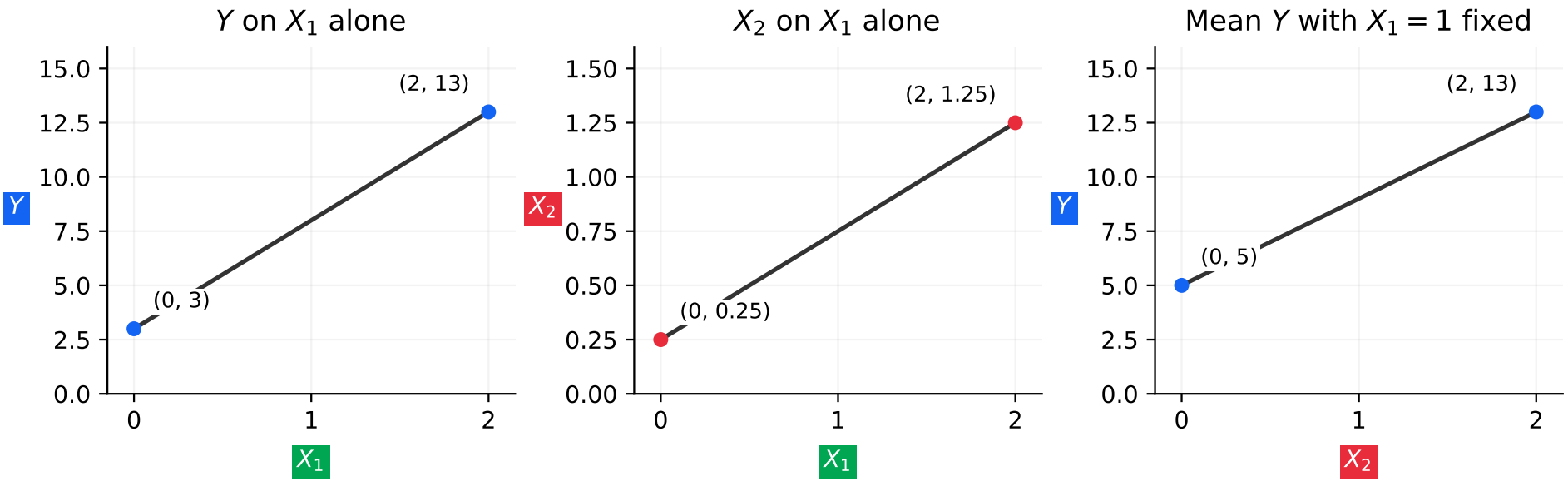
$$E[Y \mid X_{\text{van}} = 1, X_{\text{drone}} = 0] = 18 + 6(1) - 8(0) = 24 \text{ minutes.}$$

3) The drone coefficient is $\boxed{-8}$: drone deliveries average 8 minutes less than bicycle deliveries.

$$E[Y \mid X_{\text{van}} = 0, X_{\text{drone}} = 1] = 18 + 6(0) - 8(1) = 10 \text{ minutes.}$$

Recover a two-predictor model from graphs

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, with $E[\epsilon \mid X_1, X_2] = 0$. The graphs show exact population regression lines; the marked coordinates lie on those lines.



Question

Find β_0, β_1, and β_2.

Solution: coefficients from the graphs

1) The right-hand graph holds $X_1 = 1$ fixed. Its slope is the full-model coefficient on X_2:

$$\boxed{\beta_2} = \frac{13 - 5}{2 - 0} = \boxed{4}.$$

Solution: coefficients from the graphs

1) The right-hand graph holds $X_1 = 1$ fixed. Its slope is the full-model coefficient on X_2:

$$\boxed{\beta_2} = \frac{13 - 5}{2 - 0} = \boxed{4}.$$

2) Read the two simple-regression lines from the left and middle graphs:

$$\boxed{\beta_1'} = \frac{13 - 3}{2} = \boxed{5}, \qquad \boxed{\beta_{1,2}} = \frac{1.25 - 0.25}{2} = \boxed{0.5}.$$

Solution: coefficients from the graphs

1) The right-hand graph holds $X_1 = 1$ fixed. Its slope is the full-model coefficient on X_2:

$$\boxed{\beta_2} = \frac{13 - 5}{2 - 0} = \boxed{4}.$$

2) Read the two simple-regression lines from the left and middle graphs:

$$\boxed{\beta_1'} = \frac{13 - 3}{2} = \boxed{5}, \qquad \boxed{\beta_{1,2}} = \frac{1.25 - 0.25}{2} = \boxed{0.5}.$$

3) OVB gives $\boxed{5} = \beta_1 + \boxed{4}\,\boxed{0.5}$, hence $\boxed{\beta_1 = 3}$.

Solution: coefficients from the graphs

1) The right-hand graph holds $X_1 = 1$ fixed. Its slope is the full-model coefficient on X_2:

$$\beta_2 = \frac{13-5}{2-0} = 4.$$

2) Read the two simple-regression lines from the left and middle graphs:

$$\beta_1' = \frac{13-3}{2} = 5, \qquad \beta_{1,2} = \frac{1.25-0.25}{2} = 0.5.$$

3) OVB gives $5 = \beta_1 + 4\,(0.5)$, hence $\beta_1 = 3$.

4) At $X_1 = 1$ and $X_2 = 0$, the right graph gives mean $Y = 5$:

$$5 = \beta_0 + 3(1) + 4(0) \Rightarrow \beta_0 = 2.$$

Thus the model is $Y = 2 + 3X_1 + 4X_2 + \epsilon$.